trompet

A Typst package for making Tromp lambda diagrams, built on top of Lambdabus and CeTZ

github.com/CrowdingFaun624/trompet

Version 0.1.0

Guide

Reference

Tromp Diagram

There are three ways to call `tromp` to display a Tromp diagram.

```
#tromp("\\f.\\n.f (f n)")
```

```
#import "@preview/
lambdabus:0.1.0" as lmd

#tromp(lmd.parse(
  "\\f.\\n.f (f n)"
))
```

```
#tromp(expression(
  abstraction("f",
    abstraction("n",
      application(
        "f", "f n"
      )
    )
  )
))
```

All of the above produce the following Tromp diagram (the Church numeral 2):

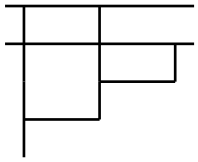
Styling

Tromp diagrams can use the `"pixel"` mode or the `"line"` mode.

`"pixel"`

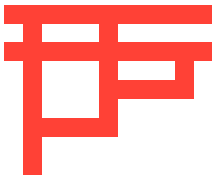
```
#tromp("\\f.\\n.f (f n)", mode: "pixel")
```

`"line"`

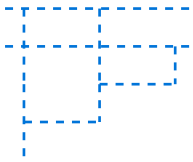


```
#tromp("\\f.\\n.f (f n)", mode: "line")
```

Tromp diagrams can adjust the styling of the fill of the pixels or the stroke of the lines using any CeTZ style.

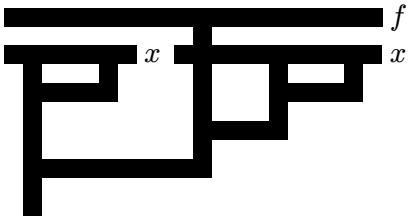


```
#tromp("\\f.\\n.f (f n)", style: red)
```

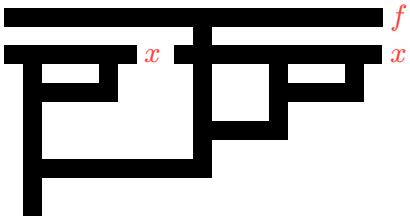


```
#tromp(
  "\\f.\\n.f (f n)",
  mode: "line",
  style: (paint: blue, dash: "dashed")
)
```

Tromp diagrams can label abstractions.



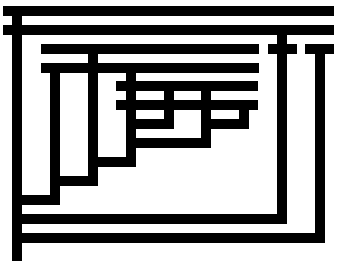

```
#tromp("λf.(λx.x x) (λx.f (x x))",
labels: auto)
```




```
#tromp(
  "λf.(λx.x x) (λx.f (x x))",
  labels: param => {
    text($italic(param)$, red)
  }
)
```

The scale of a Tromp diagram can be adjusted using the scale parameter.

```
#tromp(
  "λn.λf.n (λf.λn.n (f (λf.λx.n f (f
x)))) (λx.f) (λx.x)",
  scale: 0.5,
)
```



Expression Functions

Lambda expressions can be expressed using the `expression()`, `value()`, `abstraction()`, and `application()` functions instead of strings.

`value()`, `abstraction()`, and `application()` all take a `style` parameter which behaves the same way as above.

The `expression()` function returns an expression with additional metadata which allows Lambdabus to process it.

```
>>> #abstraction("a", value("a"))
(
  type: "abstraction",
  param: "a",
  body: (type: "value", name: "a", style:
auto),
  style: auto,
  label: auto,
))
```

```
>>> #expression(
  abstraction("a", value("a"))
)
(
  type: "abstraction",
  param: "a",
  body: (type: "value", name: "a", style:
auto, index: 0),
  style: auto,
  label: auto,
  index: 0,
)
```

The `value()` function takes in a single string for the parameter name, referring to a corresponding abstraction with the same parameter.

The `abstraction()` function takes in a string for the parameter name and a lambda expression representing the body of the abstraction. Additionally, it takes a `label` parameter which can be `auto`, `none`, a function from the parameter name to content, or just content.

The `application()` function takes in a lambda expression for the function and a lambda expression for the parameter of the application.

Functions

abstraction

Creates a stylable abstraction.

Parameters

```
abstraction(
  parameter: str,
  body: str dictionary,
  style: auto color stroke dictionary,
  label: auto none content function
)
```

parameter `str`

The string corresponding to the parameter of this abstraction.

body `str` or `dictionary`

An expression constructed through `value()`, `abstraction()`, and `application()` or a Lambdabus expression.

style `auto` or `color` or `stroke` or `dictionary`

If using the `"pixel"` mode, must be `auto` or a color. If using the `"line"` mode, must be `auto`, a stroke, or a CeTZ style. If auto, defaults to the whole diagram's default style.

Default: `auto`

label `auto` or `none` or `content` or `function`

`auto`, `none`, content, or a function that takes the parameter as a string and returns content. If `auto`, defaults to the whole diagram's label style.

Default: `auto`

application

Creates a stylable application.

Parameters

```
application(
  function: str dictionary,
  parameter: str dictionary,
  style: auto color stroke dictionary
)
```

function `str` or `dictionary`

An expression constructed through `value()`, `abstraction()`, and `application()` or a Lambdabus expression.

parameter `str` or `dictionary`

An expression constructed through `value()`, `abstraction()`, and `application()` or a Lambdabus expression.

style `auto` or `color` or `stroke` or `dictionary`

If using the `"pixel"` mode, must be `auto` or a color. If using the `"line"` mode, must be `auto`, a stroke, or a CeTZ style. If auto, defaults to the whole diagram's default style.

Default: `auto`

expression

Adds additional metadata to an expression made with `value()`, `abstraction()`, or `application()` so that Lambdabus can do math on it.

Parameters

```
expression(expression: dictionary)
```

expression `dictionary`

The lambda calculus expression constructed through `value()`, `abstraction()`, or `application()`.

tromp

Creates a Tromp diagram.

Parameters

```
tromp(
  expression: str dictionary,
  mode: str,
  scale: float,
  style: color stroke dictionary,
  labels: content auto none
) -> content
```

expression `str` or `dictionary`

A Lambdabus expression or result of `expression()`.

mode `str`

Either the string `"pixel"` or `"line"`.

Default: `"pixel"`

scale `float`

A float representing the scale from the default pixel size.

Default: `1.0`

style `color` or `stroke` or `dictionary`

If using the `"pixel"` mode, must be a color. If using the `"line"` mode, must be a stroke or a CeTZ style.

Default: `black`

labels `content` or `auto` or `none`

`auto` or a function that takes in a parameter name string and returns content.

Defaults to `param => $italic(param)$`.

Default: `none`

value

Creates a stylable value.

Parameters

```
value(
  parameter: str,
  style: auto color stroke dictionary
)
```

parameter `str`

The string corresponding to the abstraction parameter.

style `auto` or `color` or `stroke` or `dictionary`

If using the `"pixel"` mode, must be `auto` or a color. If using the `"line"` mode, must be `auto`, a stroke, or a CeTZ style. If auto, defaults to the whole diagram's default style.

Default: `auto`